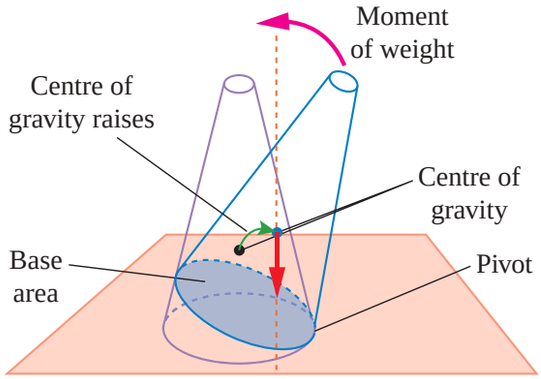
Stable Equilibrium

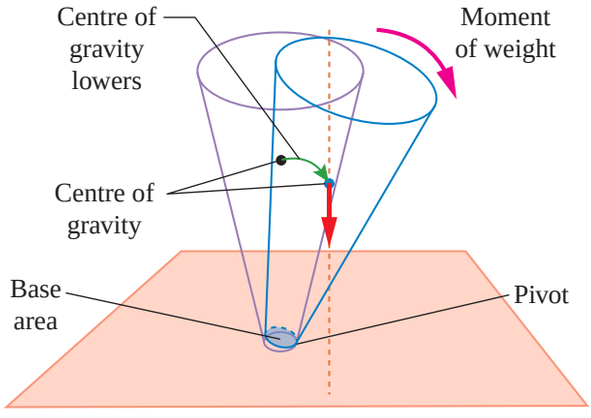
In stable equilibrium, the frustum can be tilted through quite a big angle without toppling.



Its centre of gravity is raised when it is displaced. The vertical line through its centre of gravity still falls within its base. So, it can return to its original position.

Unstable Equilibrium

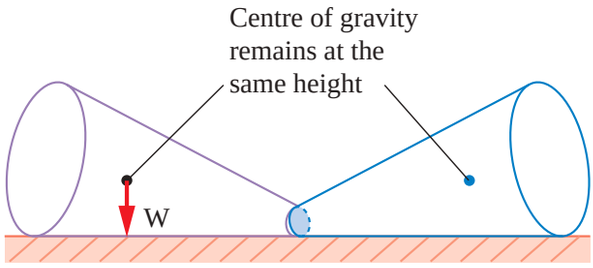
In this equilibrium, the frustum will topple with the slightest tilting. Its centre of gravity is lowered when it is displaced.



Here, the vertical line through its centre of gravity falls outside its base. So, it will not come back to its position.

Neutral Equilibrium

It causes frustum to topple. The frustum will roll about but does not topple. Its centre of gravity remains at the same height when it is displaced. The body will stay at any position to which it has been displaced.



2.7.1 Condition for Stability

Stability can be increased by the following ways.

- Lowering its centre of gravity
- Increasing the area of its base
- A heavy base lowers the centre of gravity So, the object will be stable.
- A broad base makes the object more stable.



The Thanjavur Doll

It is s type of traditional toy made in Thanjavur from terracotta material. The centre of gravity and the total weight of the doll is concentrated at its bottom most point, generating a dance-like continuous movement with slow oscillations.



2.7.2 Real Life Applications of Centre of Gravity

- In order to have stability, the luggage compartment of a tour bus is located at the bottom and not on the roof.
- Extra passengers are not allowed on the upper deck of a crowded double decker bus.

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